



OFFICIAL

Perioperative Fasting and Medication Management Guideline

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1. Purpose

To ensure that adult patients are fasted appropriately and receive optimal clinical management of their medications prior to their surgical or endoscopic procedure.

For paediatric patients refer to the [Fasting the Child for Anaesthesia Guideline](#).

2. Background

Some of the regular medications taken by a patient might need to be modified or discontinued in the peri-operative period to provide optimal conditions for anaesthesia and/or surgery. Other medications must be continued to reduce the risk of complications peri-operatively.

Gastric emptying of clear fluids occurs within two hours after ingestion. Therefore, patients can take medications with small amounts of water prior to surgery. Medication that must be taken with food should be withheld on the day of surgery/ procedure, unless instructed by a doctor to continue despite fasting.

The recommendations in this guideline are based on available references and on clinical experience.

This document does not cover all clinical scenarios, and it must be individualised for each patient, after discussion with the anaesthetist and/or surgeon as required.

3. Fasting

Type of food/drink	Fasting period recommended	Comments
Clear fluids	2 hours	Drinks through which newsprint can be read, including black tea and coffee, carbonated drinks, carbohydrate drinks, lemonade, clear fruit juice (without pulp) or cordial, isotonic drink. *Note a carbohydrate drink may be required for carbohydrate loading in some major surgeries. Soups, yoghurt, particulate drinks, smoothies or milk-containing drinks and jellies do not fall in this category.
Breast milk	4 hours	
Formula and all other types of milk	6 hours	Tea and coffee with milk are acceptable up to 6 hours before induction of anaesthesia.
Light meals, lollies and sweets	6 hours	For example, toast and a clear drink. Fried or fatty foods should be avoided as they can prolong gastric emptying

Type of food/drink	Fasting period recommended	Comments
Fried foods, fatty foods, meat	8 hours	This longer period may be required at anaesthetic clinician's discretion
Cigarettes, vaping and chewing gum	Nil extra time	While their use in the pre-operative period is strongly discouraged and they all increase gastric content, there is no evidence that increased fasting is required

4. Fasting and elective surgery

Elective patients must be fasted before surgery or non-surgical procedures which involve regional or general anaesthesia or sedation in theatres, or procedures requiring sedation outside theatres. These include electro-convulsive therapy, gastro-intestinal endoscopies, DC cardioversions and radiological procedures requiring sedation or anaesthesia.

Patients should be encouraged to drink clear fluids up to 200 ml/hour till 2 hours before surgery (up to 6am for morning lists and up to 11am for afternoon lists) as it improves patient well-being. This includes pregnant women not in labour.

Elective surgery should not be undertaken if the patient does not satisfy the above fasting criteria.

High risk patients, such as those who are obese, pregnant, have gastro-oesophageal reflux disease, gastric band in-situ or diabetics, should be treated with appropriate pre-medication, such as ranitidine, sodium citrate, metoclopramide, or proton-pump inhibitors.

Regular medication should be continued (hospital protocol should be followed for antiplatelet, anticoagulants, oral hypoglycaemic medications, and insulin) with up to 30mL of water (0.5mL / kg in paediatric patients). Antihypertensive drugs should not be stopped unless specific orders are made to do so. Any medications normally administered with food, including non-steroidal anti-inflammatory drugs (NSAIDs) and iron should be withheld during the fasting period, except steroids, which should not be stopped.

In case of an unplanned delay in the start of elective surgery, consideration should be given to giving the patient a drink of water to prevent thirst and dehydration, after consultation with the anaesthetist.

5. Fasting and emergency surgery

The urgency of any emergency surgery should be discussed with the surgeon, and fasting guidelines should be followed wherever possible.

It should be remembered that this is still not likely to render an emergency patient "fasted" as in an elective surgery, due to various factors such as anxiety, pain, trauma, opioid administration, nausea and vomiting, gastro-intestinal obstruction etc., all of which should be taken into consideration by the anaesthetist while making a decision about the type of induction and airway management.

Patients posted for emergency surgery should be ideally on intra-venous fluids, as the timings are more uncertain, and these patients are at a higher risk of dehydration.

6. Medication management

This procedure covers the following categories of medications:

1. [Cardiovascular medications](#)
2. [Anticoagulants](#)
3. [Antiplatelets](#)
4. [Nonsteroidal anti-inflammatory agents](#)
5. [Diabetic medications](#)
6. [GLP IRAs and GLP-1GIPRAs](#)
7. [Neuropsychiatric medications](#)
8. [Opioid analgesics](#)
9. [Proton pump inhibitors and H₂ antagonists](#)
10. [Immunosuppressants](#)
11. [Oral contraceptives and hormone replacement therapy](#)
12. [Non-prescription and herbal medications](#)
13. [Medical marijuana](#)

6.1 Cardiovascular medications

Antihypertensives

Patients **must not** withhold antihypertensive medications pre-operatively.

AM procedures:

Unless specified continue morning medications. Take medication with a sip of water before coming into hospital.

PM procedures:

Unless specified continue morning medication with their breakfast.

Hypolipidemics - Continue as usual

Antiarrhythmics - Continue as usual

Diuretics - Usually stopped on morning of surgery; based on patient assessment

6.2 Anticoagulants

Oral anticoagulant therapy is used to decrease the thromboembolic complications of

- Atrial fibrillation
- Venous thromboembolism
- Mechanical heart valves

However, it can place the patient at risk of peri-operative bleeding related to surgical or anaesthetic procedures.

Warfarin is a commonly used oral anti-coagulant. Other new oral anticoagulants are now being used increasingly and they include:

- Dabigatran (Pradaxa)
- Rivaroxaban (Xarelto)

- Apixaban (Eliquis)

The peri-operative care of these patients aims to minimize the risks of thromboembolism and bleeding by the following strategies.

- Identifying patients at risk, the indication(s) and anticoagulant used
- Assessing the thromboembolic risk
- Assessing the procedure related bleeding risk
- Providing clear instructions with regard to
 - discontinuing anticoagulants
 - timing of discontinuation
 - bridging therapy when needed.
 - recommencement of therapy

Pre-operative management of anticoagulant medications depends on the indication for anticoagulation and the risk of thrombosis when the drug is withheld relative to the risk of bleeding if continued.

Advanced planning for bridging therapy, reversal strategies and management of bleeding during and after surgery must be considered. Advice from a cardiologist and/or haematologist may be necessary in complex situations.

It should be noted that the net clinical benefit of bridging patients at intermediate to high thromboembolic risk is not yet supported by clinical trial evidence, but excessive haemorrhagic events have been well described.

The recent BRIDGE trial shows that in patients on warfarin for atrial fibrillation (AF), not bridging is non-inferior to bridging for preventing thromboembolism and reducing bleeding when stopping warfarin. Therefore, when warfarin is prescribed only for AF, and is to be withheld prior to surgery, bridging with heparin is not recommended.

The proceduralist will:

- Decide on a peri-operative plan with input from consultants of other specialties managing the anticoagulation (e.g., cardiologist, haematologist, general practitioner, anaesthetist) if needed
- Decide if it is safe to continue anticoagulation till the day before surgery
- Document in, or attach this plan to MR20
- Oversee the recommencement of anticoagulants post procedure

The preadmission clinic nurse will:

- Ask the patient if they take medications that thin their blood, as per the 'Preadmission Clinic Patient Assessment Package', and document the generic names of these medications on the AKMR 93 (Nursing care plan)
- Check the patient's last INR result (for those on warfarin)
- Discuss all patients undergoing elective surgery on anticoagulation (seen only by nurse) with the anaesthetist
- Organise an INR on the day of surgery/procedure (AM patients on the ward via early morning collection and PM patients to come in 30min early and attend Pathology).

The anaesthetist will:

- Determine the indication(s) for anticoagulant
- Assess the thromboembolic and bleeding risk (Tables 1 and 2)
- Initiate further discussion with other specialties or GP as deemed necessary. e.g. Obtain and document advice from the patient's cardiologist regarding appropriate bridging anticoagulation in patients with mechanical heart valves
- Ensure that there is a peri-operative plan and where necessary discuss with surgeon/proceduralist / anaesthetist regarding discontinuation, timing and need for bridging therapy
- Involve the GP if required if they need to:
 - Arrange cessation of anticoagulants prior to surgery.
 - Commence the patient on bridging enoxaparin if required

For patients undergoing procedures in the endoscopy suite:

- Patient(s) on anticoagulation undergoing endoscopies shall have their anticoagulation managed by their GP or referring doctor. The patient will be provided with a 'Guidelines for endoscopy patients on warfarin / anticoagulation' sheet if their anticoagulant medication needs to be withheld pre-operatively.

Table 1: Suggested risk stratification for procedural bleeding risk

High bleeding risk procedures (2-day risk of major bleed of $\geq 2\%$)	Low bleeding risk procedures (2-day risk of major bleed of $< 2\%$)	Minimal bleeding risk procedures
• Major surgery (procedure duration of > 45 min) • Cancer surgery • Major orthopaedic surgery • Urologic, including transurethral prostate resection, bladder resection, tumour ablation, nephrectomy, kidney biopsy. • Gastrointestinal surgery including bowel resection • Colonic polyp resection • Surgery in highly vascular organs (kidneys, liver, spleen) • Spinal or epidural anaesthesia	• Arthroscopy • Cutaneous/lymph node biopsies • Shoulder/foot/hand surgery • Coronary angiography • Gastrointestinal endoscopy/ colonoscopy $\pm$ biopsy • Abdominal hysterectomy • Laparoscopic cholecystectomy • Abdominal hernia repair • Haemorrhoid surgery • Epidural injections with INR of < 1.2	• Minor dermatologic procedures (excision of basal and squamous cell skin cancers, premalignant or cancerous skin naevi) • Pacemaker or cardioverter-defibrillator device implantation* • Minor dental procedures (dental extractions, restorations, prosthetics, endodontics), dental cleanings, fillings • Cataract surgery

INR- International Normalised Ratio.

*Associated with pocket hematoma but randomized controlled trial (Level 1) evidence reveals that procedures can be performed without oral anticoagulant interruption.

Table 2: Suggested patient-related risk stratification for peri-operative thromboembolism from American College of Chest Physicians antithrombotic guidelines

Risk category	Mechanical heart valve	Atrial fibrillation	Venous thromboembolism
High (> 10% per year risk of ATE or > 10% per month risk of VTE) Anticoagulation advised	- Any mechanical mitral valve - Caged ball or tilting disk valve in mitral/aortic position - Recent (< 6 months) stroke or TIA	- CHADS₂ score of 5 or 6 - Recent (< 3 months) stroke or TIA - Rheumatic valvular heart disease	- Recent (< 3 months) VTE - Severe thrombophilia - Deficiency of protein C, protein S, or antithrombin - Antiphospholipid antibodies - Multiple thrombophilias
Intermediate (4–10% per year risk of ATE or 4–10% per month risk of VTE) Anticoagulation advised on a case-by-case basis	Bileaflet mechanical aortic valve with major risk factors for stroke	CHADS₂ score of 3 or 4	- VTE within the past 3–12 months - Recurrent VTE - Non-severe thrombophilia - Active cancer
Low (< 4% per year risk of ATE or < 2% per month risk of VTE) Anticoagulation usually not advised	Bileaflet mechanical aortic valve without major risk factors for stroke	CHADS₂ score of 0–2 (and no prior stroke or TIA)	VTE more than 12 months ago

- ATE, arterial thromboembolism; TIA, transient ischemic attack; VTE, venous thromboembolism.

Table 3: CHADS₂ scoring

CHAD ₂ acronym	Score
Congestive HF	1
Hypertension	1

Age ≥75 years	1
Diabetes mellitus	1
Stroke/TIA/TE	2

HF: heart failure; MI: myocardial infarction; TE: thromboembolic; TIA: transient ischemic attack.

6.2.1 Warfarin

Table 4: Suggested peri-procedural anticoagulant and bridging management for patients on long-term oral anticoagulants (including vitamin K antagonists and direct oral anticoagulants) based on thromboembolic risk and procedural bleeding risk			
	High bleeding risk procedures	Low bleeding risk procedures	Minimal bleeding risk procedures
High thromboembolic risk (on warfarin)	Interrupt warfarin therapy with LMWH bridging suggested on the basis of clinician judgement and the most current evidence *Θ	Interrupt warfarin therapy with LMWH bridging suggested on the basis of based on clinician judgement and the most current evidence *	Do not interrupt anticoagulant therapy
Intermediate thromboembolic risk (on warfarin)	Consider interrupting warfarin therapy without LMWH bridging on the basis of clinician judgement and the most current evidence *Θ	Consider interrupting warfarin therapy without LMWH bridging on the basis of clinician judgement and the most current evidence *	Do not interrupt anticoagulant therapy
Low thromboembolic risk (on warfarin)	Interrupt warfarin therapy; bridging with LMWH not necessary Θ	Interrupt warfarin therapy; bridging with LMWH not necessary	Do not interrupt anticoagulant therapy
DOAC users	DOAC therapy may be interrupted without the need for bridging with LMWH.		Do not interrupt DOAC anticoagulant therapy. May consider interrupting DOAC therapy on the day of the procedure.

LMWH- low molecular weight heparin

DOAC- direct oral anticoagulant

* Atrial fibrillation: bridging is not recommended on the basis of Level 1 evidence, but evidence exists in a few high-risk CHADS₂ patients (scores of 5 and 6).

⊖ May administer prophylactic-dose of LMWH for VTE prevention in patients undergoing high bleeding risk procedures or major surgeries that confer a high risk of VTE.

DOACs (Directly acting oral anticoagulants)

(A) Factor Xa inhibitors: Rivaroxaban, Apixaban, Edoxaban, Betrixaban

It is recommended that these drugs should be discontinued 72 hours prior to neuraxial block. (Consider checking drug levels or anti-factor Xa level if less than 3 days.)

(B) Direct thrombin inhibitor: Dabigatran

It is recommended that these drugs should be discontinued 120 hours prior to neuraxial block. Recent evidence states that in patients with no other risk factors for bleeding such as age > 65 years, hypertension, concomitant use of antiplatelets, with a reliably determined renal function, a more graded approach can be followed, as described in the following table:

	Dabigatran		Rivaroxaban		Apixaban	
	Surgical Bleeding Risk					
	Low risk	High risk	Low risk	High risk	Low risk	High risk
Renal Function	Last dose before surgery					
CrCl >50 mL/min	48 hours	72 hours	48 hours	72 hours	36 hours	48-72 hours
CrCl 30-49 mL/min	72 hours	120 hours	48 hours	72 hours	48 hours	72 hours
Urgent surgical intervention	• If possible delay surgery until at least 12-24 hours after last dose of DOAC • Assess increased risk of bleeding against urgency of surgery • If surgery cannot be delayed seek haematologist advice					

6.2.2 Heparin

Anticoagulant drug	Recommended interval to block
IV heparin	6 hr
Unfractionated s/c heparin, prophylactic dose	6 hr
Unfractionated s/c heparin, prophylactic dose > 15000 U/day	12 hr
Unfractionated s/c heparin, therapeutic dose	24 hr
LMWH	
Enoxaparin (prophylactic) (20 or 40 mg od)	12 hr

Enoxaparin (therapeutic)(1mg/kg/bd or 1.5mg/kg daily)	24 hr
Dalteparin	24 hr

*Since patients on heparin (unfractionated or LMW) can develop thrombocytopenia, it is recommended that patients on heparin for more than 4 days have their platelet count checked prior to a neuraxial block.

6.3 Antiplatelets

Cessation of any anti-platelet therapy should never be considered routine; there are patients in whom the thrombotic/ischaemic risks outweigh the risk of major haemorrhage.

Individual patient assessment is essential, and those on dual antiplatelet therapy or those with coronary stenting **must not** stop their antiplatelet drugs without receiving advice from their cardiologist.

- **Aspirin:** Due to its cardio-protective effects, aspirin should be continued in the pre-operative period (excluding on the day of surgery as they will be on empty stomach) in all patients, especially those with coronary stents. It may be withheld for 7 days before surgery in patients who are at a very high risk of bleeding or if the consequences of bleeding are significant.
- Patients should be assessed on an individual basis and discussed with specialists if necessary. Instruction/s from the doctor must be documented in the medical notes and on the medication chart.
- **Clopidogrel:** This is a more potent antiplatelet agent than aspirin but with few studies quantifying its effects on post-operative bleeding. In most cases, it is recommended to withhold clopidogrel for 7 days before surgery.
- In patients with coronary stents, individual assessment of the risk of bleeding versus the risk of stent thrombosis if stopping clopidogrel prematurely is required. This is especially important in patients who have had a drug eluting coronary stent inserted in the past 12 months or a bare metal stent in the past 2-4 weeks.
- Note: Combination products of clopidogrel and aspirin are widely used. Patients should understand instructions clearly if they are instructed to stop clopidogrel only but to continue aspirin.
- **Dipyridamole:** Limited data exist on the safety of continuing dipyridamole in the peri-operative period. Consult the anaesthetist and surgeon before withholding dipyridamole and cease at least 2 days before surgery.
- It is often prescribed in conjunction with aspirin, in which case the combination should be ceased 7 days before surgery.
- **Prasugrel:** Stop 7 days before surgery after consultation with cardiologist.
- **Ticagrelor:** Stop 5 days before surgery after consultation with cardiologist.
- **Ticlopidine:** Stop 14 days before surgery after consultation with cardiologist.
- Platelet Glycoprotein IIb/IIIa inhibitors (e.g., abciximab, eptifibatide and tirofiban) - Must consult cardiologist before stopping.

The anaesthetist will:

- Establish the balance of risks for each patient in conjunction with other disciplines (proceduralist, cardiologist, haematologist) where needed; and contact the treating cardiologist in the presence of coronary stents and document their advice
- Provide a plan for continuation or cessation of agents to the patient and the pre-admission clinic
- For patients with coronary stents in-situ (bare metal or drug-eluting), dual antiplatelets must not be stopped without prior discussion with treating cardiologist. Where surgery cannot be postponed, it may need to be performed at a tertiary hospital with Percutaneous Coronary Intervention (PCI) facilities

The preadmission clinic nurse will:

- Ask the patient if they take aspirin and/or other anti-platelets, as per the 'Preadmission Clinic Patient Assessment Package'
- Record the medication(s) and its generic name(s)
- Assess the indication(s) and whether the patient has undergone coronary stenting in the past (for patients seen only by nurse), and discuss those deemed high risk with the PAC anaesthetist
- Obtain relevant cardiac investigation reports/correspondence from their healthcare providers for patients with significant cardiac history
- Consult the anaesthetist for a plan regarding antiplatelet medications
- Document all instructions received carefully regarding anti-platelet medications on the AKMR93 and ensure the patient understands the instructions

6.4 Nonsteroidal anti-inflammatory drugs

These include celecoxib, meloxicam, diclofenac, ibuprofen, indomethacin, ketoprofen, naproxen.

Non-selective nonsteroidal anti-inflammatories (NSAIDs) such as ibuprofen and diclofenac have reversible antiplatelet effects whereas COX-2 inhibitors such as celecoxib have minimal effects on platelet function. All NSAIDs may cause renal toxicity.

NSAIDs should be withheld on day of surgery/procedure.

6.5 Diabetes mellitus

Patients with diabetes mellitus requiring surgery have a higher risk of morbidity and mortality in the peri-operative period, and they need specialist care during the peri-operative period for their glycaemic control. Poor pre or peri-operative blood glucose control influences the risk of post-operative complications including wound infection.

These guidelines will address some of the requirements for optimal diabetes management. Blood glucose monitoring should be performed pre-, intra-, and post-operatively throughout the patient's journey at regular intervals and any hypoglycaemia or hyperglycaemia reported to the patient's anaesthetist for management.

This procedure should be used only for the management of diabetic patients.

The preadmission clinic nurse will:

- Assess the patient for a history of diabetes mellitus as per the 'Preadmission Clinic Patient Assessment Package' and document on the AKMR93 and on the one-note handover to theatre.

- Ask if the patient has Type 1 or Type 2 diabetes mellitus, if the patient is on insulin therapy or not (all patients treated with insulin should be managed in the same way, irrespective of the type of diabetes), and perform a blood sugar level reading for patients seen in person in PAC.
- Note the patient's diabetic medications and insulin, with dosages and timings for nurse-only patients, and get advice from the anaesthetist about pre-operative management.
- Ensure HbA1C is checked for elective surgery. (HbA1C>9 is indicative of high surgical risk for elective surgery; these should be postponed if possible and reviewed by endocrinology).
- Discuss standard fasting instructions for AM and PM surgery.
- Provide the following medication instructions (Please refer to tables 5,6 and 7 for instructions regarding simple anti-diabetic medications; please liaise with the PAC anaesthetist if there are any concerns)

Table 5: Commonly used hypoglycaemic agents		
Oral hypoglycaemic agents (OHAs) - all except SGLT2 inhibitors	Sulphonylureas Metformin* DPP-4 inhibitors: linagliptin, sitagliptin, vildagliptin etc Glitazones Meglitinides Acarbose	Ceased on day of surgery and restarted once oral intake is adequate
Oral hypoglycaemic agents (OHAs)- SGLT2 inhibitors #	Canagliflozin (Invokana - available in New Zealand but not in Australia) Dapagliflozin (Forxiga) Empagliflozin (Jardiance) In combination with metformin (Xigduo, Jardiamet)	-refer to Table 7-
Incretin mimetics	Liraglutide (Victoza) Dulaglutide Semaglutide Tirzepatide	-refer to section 6.6-
Basal insulin	NPH Glargine (Lantus) Determir (Levemir)	-refer to Table 6-
Bolus meal- time insulin	Fast/ rapid-acting insulin (e.g. Novorapid, Humalog, Apidra, Actrapid, Humulin R)	- refer to Table 6-
Premixed insulin	Mixture of intermediate and fast/rapid-acting insulin (e.g. Novomix, Humalog Mix 25, Mixtard 30/70)	- refer to Table 6-

*Metformin- see Special Instructions Table (table 7) - can be continued throughout fasting period in minor surgery.

SGLT2 inhibitors- see separate instructions in special instructions table (table 7). Instructions apply regardless of indication of the medication (diabetes or cardiac).

- Instruct patients to bring all their diabetes medications with them to the hospital to ensure accurate identification of the type of insulin or oral anti-hyperglycaemic medications by medical and nursing staff and to facilitate resumption of their usual therapy post-operatively.
- Ensure that the patient is capable of managing their diabetes after discharge from hospital, for example, has a support person at home, and is physically and mentally capable of checking BGLs and administering insulin. Instruct patient to involve their GP in case of any long-term concerns.
- Inform theatre that the patient is on insulin by completing a hand over form in preadmission. Patients with diabetes should be preferably on the morning list, and preferably first on the list.
- Remind patients to monitor their blood sugars and that if they become hypoglycaemic and are still at home, they should have three teaspoons of sugar dissolved in warm water. This may delay their procedure, but it is unlikely to be cancelled. Patients should ensure they are well hydrated before the procedure. Due to the possibility of hypoglycaemia whilst fasting, patients should preferably not drive themselves to the hospital on the day of surgery.
- **TARGET:** BGL should be kept between 4-12mmol/l during the peri-operative period. BGL should be monitored 2-hourly for patients not on insulin, and 1-hourly for patients on insulin or those on insulin-dextrose infusion.
- Patients that experience hypoglycaemia on the day of surgery should be given 60ml of CARBOTEST drink after informing the relevant anaesthetist {or the Duty Anaesthetist (DA) if uncontactable}. If the patient is symptomatic for hypoglycaemia, the anaesthetist doing the list or, if not contactable, the DA can be informed after administration of Carbotest by the nurse. This may delay surgery by 2 hours but is unlikely to result in cancellation of the surgery. Patients who are hyperglycaemic should be treated with an insulin-dextrose infusion in the peri-operative period according to Protocol AKMR174. The responsible anaesthetist, or the DA, must be informed if the BGL < 4 or > 12mmol/l.

For the purpose of this document, minor surgery is defined as all same-day-only procedures, while major surgery includes all procedures that require at least an overnight admission.

Note: "Sliding scale insulin" is **NOT** recommended for the peri-operative management of diabetes when used as sole therapy.

Table 6: Management of patients on insulin

Patients on insulin		
Minor surgery	Morning List	• Continue normal insulin dose, OHAs and diet until the day before surgery. (For patients with morning BGLs < 5mmol/l, decrease long-acting/ basal doses by 10% on the night before surgery). Fast from midnight. • Omit morning insulin (long as well as short-acting) and OHAs.* • If possible, the patient may have a late breakfast after surgery with their regular dose of insulin. • BGL monitoring at admission and hourly thereafter. • For later procedures, or if the BGLs are erratic, consider starting insulin-dextrose infusion.

Table 6: Management of patients on insulin

Patients on insulin		
	Afternoon list (Afternoon procedures are not ideal for insulin-treated patients as they are more disruptive to their glycaemic control)	- Continue normal insulin dose, OHAs and diet until the day before surgery. - Omit OHAs on morning of surgery.* - Long acting insulin- usual dose if taken in the evening, half the usual dose before breakfast if taken in the morning. - Rapid-, short- or intermediate-acting insulin (including premix) - if taken in the morning, take half the usual dose on the morning of surgery with a light breakfast (finished by 7 am). - BGL monitoring at admission and hourly thereafter. - Consider starting insulin-dextrose infusion if BGLs are high/erratic. - Restart normal oral intake at dinner, with usual dose of evening insulin and OHAs.
Major surgery	Morning List	- Continue normal insulin dose and diet till day before surgery. (For patients with morning BGLs < 5mmol/l, decrease long-acting/basal doses by 10% on the night before surgery). Fast from midnight. - Omit morning insulin (long as well as short-acting) and OHA. * - Consider commencing an insulin-dextrose infusion prior to induction of anaesthesia (or by 1000hrs at the latest). - BGL measurements at admission and hourly thereafter. - Post-op, continue insulin-dextrose infusion until adequate oral intake; then stop the infusion and restart OHAs and insulin. Check renal function before recommencing metformin after a major surgery.
	Afternoon list (Afternoon procedures are not ideal for insulin-treated patients as they are more disruptive to their glycaemic control)	- Continue normal insulin dose, OHAs and diet till day before surgery. - Omit OHAs on morning of surgery. * - Long acting insulin- usual dose if taken in the evening, half the usual dose before breakfast if taken in the morning. - Rapid-, short- or intermediate-acting insulin (including premix) - if taken in the morning, take half the usual dose on the morning of surgery with a light breakfast (finished by 7 am). - BGL measurements on admission hourly thereafter. - Post-operatively, restart normal insulin regime and OHAs with adequate oral intake usually at dinner, or continue insulin-dextrose infusion with hourly BGL monitoring till oral intake is adequate. - Check electrolytes daily and check renal function before recommencing metformin after a major surgery.
Patients not on insulin (major/minor; morning/afternoon)		- Continue as usual until the day before surgery. - Omit OHAs on the morning of surgery.* - BGL monitoring 2 hourly from waking (at home and after arrival at hospital).

Table 6: Management of patients on insulin

Patients on insulin	
	• Use insulin dextrose infusion and monitor BGLs hourly, if: 1. if BGL>12mmol/l or 2. surgery is prolonged and complicated or 3. Consider in major surgery in a patient on more than one antidiabetic agent. • Return to normal regime once oral intake is adequate, and continue 2 hourly BGL monitoring until stable. • After major surgery, check renal function before commencing metformin.
Emergency and time-critical surgery	• Involve the DA in decision making. • Commence on insulin-dextrose infusion if persistently hyperglycaemic. • Liaise with medicine/endocrinology about management of complex cases. • Liaise with medicine/endocrinology if there is any concern regarding diabetic ketoacidosis or hyperosmolar state.

Metformin: see separate instructions in special instructions table (table 7) → can be continued throughout fasting period in minor surgery.

Table 7: Special circumstances

Bowel preparation For patients who are not seen in the PAC in person, it is important for them to see their GP to have a diabetic medication management plan arranged.	The doses of insulin and OHAs need to be modified for patients on clear fluids for bowel preparation. • Stop the short acting insulin and reduce the pre-mixed insulin doses to half while the patient is on clear fluids. • Patients who take long-acting (e.g. Lantus) in combination with short-acting insulin may continue their long-acting insulin, while omitting the short-acting one. • Regular BGL monitoring is necessary (2 hourly if unstable glycaemic control; 4 hourly if stable) • “Clear fluids” should contain some glucose, and not just “diet drinks” or “diet jelly”. Drinks containing added sugar should be consumed if BGL falls below 5mmol/l. Conversely, drinks containing less sugar (broths or diet drinks) should be consumed if BGL exceeds 10mmol/l. • Consider admission during the clear fluids stage with dextrose-insulin infusion in poorly controlled patients.
Radiological contrast Patients on metformin with impaired renal function should	• Requires peri-operative evaluation. • Low risk if normal renal function, minor surgery, small fluid shifts.

follow up with their GP to ensure that renal function is stable.	The requesting clinician should provide the most recent serum creatinine level to the radiologist. Metformin may be omitted on day of study.
Metformin -see above-	- If there is significant renal impairment, withhold metformin on the day of surgery, but consider an alternative method of glycaemic control. - Minor surgical patients can continue metformin through the fasting period. - Consider checking renal function before recommencing metformin after major surgery and after radio-contrast administration.
Insulin pumps	- For patients undergoing minor surgery, continue pumps at basal rates and monitor BGLs hourly, changing over to dextrose-insulin infusion if required. - For patients undergoing major or emergency surgery, turn the pump off and replace it with an insulin-dextrose infusion. - Liaise with endocrinology/medicine team for pump management.
SGLT2 Inhibitors	In diabetics on SGLT2is: For day-stay procedures (including gastroscopy), these drugs can be stopped just for the day of procedure. However, fasting before and after the procedure should be minimised. - For surgery/procedures that require hospitalization, bowel preparation or carbohydrate restriction (e.g., colonoscopy), these should be ceased 3 days pre-operatively (2 days prior to surgery and the day of surgery). This may require an increase in other glucose-lowering drugs during that time. In non-diabetics on SGLT2is: SGLT2is are also being used for non-diabetic heart failure and renal failure patients, to delay progression of these diseases. These patients should have their SGLT2i medication withheld for 3 days to reduce the risk of euglycemic ketoacidosis. - Strongly consider postponing non-urgent surgery if SGLT2 inhibitors have not been ceased 3 days prior to surgery, and blood ketones are >1.0mmol/L in the perioperative period, or where HbA1c is >9.0%. - Consider replacing combination antidiabetic medications containing SGLT2Is, e.g. with metformin, with metformin alone. You could write a prescription for metformin for 3 days or forward this request to the GP.

- It is important to provide transition care until the patient is able to tolerate adequate oral intake and can return to their normal regime.
- Ensure that adequate antiemetic therapy is written up.
- Ensure BGLs are recorded 2 hourly until BGL is stable and the patient is on normal regime.
- Check serum electrolytes daily for patients having major surgery or on intravenous insulin glucose infusion.

- Tailor the guidelines to meet patient requirements in major surgery or in very sick patients.
- Ensure adequate education is provided to the patient and liaise with the medicine/endocrine team, if required, to facilitate a safe transition to appropriate regime.
- In day-case surgery, ensure a support person is with the patient for 24 hours who is aware of hypoglycaemia monitoring.

6.6 GLP IRAs and GLP-1/GIPRAs

Table 8

Agent	Receptor agonism	Elimination half-life	Administration schedule	Trade name
Liraglutide	GLP-1	12.6 - 14.3 hours	Once daily	Victoza (up to 1.8 mg) Saxenda (up to 3 mg)
Dulaglutide	GLP-1	4.7 - 5.5 days	Once weekly	Trulicity
Semaglutide	GLP-1	5.7 - 6.7 days	Once weekly	Ozempic (up to 1 mg) Wegovy (up to 2.4 mg)
Tirzepatide	GLP-1 & GIP	4.2 - 6.1 days	Once weekly	Mounjaro

- All patients should be asked about the use of GLP-1RAs and GLP-1/GIPRAs prior to anaesthesia or sedation for surgical and endoscopic procedures.
- Elective preprocedural cessation of GLP-1RAs and GLP-1/GIPRAs is not recommended.
- Patients should be asked about the use of other medications and medical conditions which may delay gastric emptying, such as bowel dysmotility, gastroparesis, and Parkinson's disease.
- All patients on these drugs should receive the two information sheets created by ANZCA, titled "PATIENT INFORMATION Preparing for your medical procedure when taking some diabetes and/or weight loss medications"
- Preprocedural diet modification with 24-hr clear fluid diet (similar to colonoscopy procedures), followed by restricted fluid intake commencing 6 hours preoperatively, and then the standard preoperative fluid fast from 2hrs preoperatively, should be recommended for all patients receiving GLP-1 RAs and GLP-1/GIPRAs.
-

On the whole day before, the patient should not eat any solid food and should drink "clear fluids" only.

	To stop clear fluids except water	To drink only water (125 ml/half cup every hour)	To stop all drinks
Am procedure/surgery	Midnight	Midnight-6 am	After 6 am
Pm procedure/surgery	6 am	6 am- 11 am	After 11 am

6.7 Neuropsychiatric medications:

Antipsychotics, antidepressants, mood stabilizers, anti-epileptics and anti-parkinsonians should be continued as usual. A post-operative medication management plan must be devised prior to

surgery, in consultation with a specialist if necessary. Potential drug interactions with other drugs used during surgery should be considered.

6.8 Opioid analgesics:

It is important not to withhold patients' regular controlled-release analgesics including transdermal patches used for chronic pain prior to surgery unless indicated by the anaesthetist. Post-operatively, the anaesthetist will prescribe an analgesic regime according to individual patient's requirement.

Regular controlled-release analgesics are usually continued but should be reviewed post-operatively and the decision to continue or withhold regular controlled-release analgesia should be clearly documented on the prescription by the anaesthetist or medical officer.

6.9 Proton-pump inhibitors and H2-antagonists:

For example: esomeprazole, lansoprazole, omeprazole, pantoprazole, rabeprazole, famotidine, nizatidine, ranitidine.

It is important to continue these medications until the morning of surgery, especially if they are being taken for reflux.

6.10 Immunosuppressants

- **Steroids:** Patients taking long-term, high dose steroids (greater than 5mg/day of prednisolone or its equivalent for 4 weeks or longer) have an impaired natural stress response and discontinuation of the steroids pre-operatively may precipitate adrenal insufficiency, increasing the likelihood of circulatory collapse and shock if adequate replacement therapy is not provided. Patients who formerly have been taking regular steroids should be treated as if on steroids, if it has been less than 3 months since stopping steroids.
- **Disease-modifying anti-rheumatic drugs:** e.g., azathioprine, hydroxychloroquine, leflunomide, methotrexate, sulfasalazine: Limited literature, continue as usual.
- **Biologic agents:** e.g., adalimumab, infliximab, certolizumab pegol, golimumab, anakinra, abatacept, rituximab, tocilizumab, tofacitinib, baricitinib: Limited literature, continue as usual.

Table 9

	Intra-operative steroid replacement	Postoperative steroid replacement
Major surgery	Hydrocortisone 100 mg intravenously at induction, followed by immediate initiation of a continuous infusion of hydrocortisone at 200 mg.24 h ⁻¹ ; Alternatively, dexamethasone 6–8 mg intravenously, if used, will suffice for 24 h	Hydrocortisone 100 mg.24 h ⁻¹ by i.v. infusion while nil by mouth (alternatively, hydrocortisone 50 mg every 6 h by i.m. injection) Resume enteral glucocorticoid at pre-surgical therapeutic dose if recovery is uncomplicated. Otherwise continue double oral dose for up to a week
Body surface and intermediate surgery	Hydrocortisone 100 mg, intravenously at induction, followed by immediate initiation of a continuous infusion of hydrocortisone 200 mg.24 h ⁻¹ Alternatively, dexamethasone 6–8 mg intravenously, if used, will suffice for 24 h	Double regular glucocorticoid dose for 48 h, then continue usual treatment dose if uncomplicated
Bowel procedures requiring laxatives/enema	Continue normal glucocorticoid dose. Equivalent i.v. dose if prolonged nil by mouth Treat as per primary adrenal insufficiency if concerned about hypothalamo-pituitary-adrenal axis function, and risk of adrenal insufficiency	
Labour and vaginal delivery	Hydrocortisone 100 mg intravenously at onset of labour, followed by immediate initiation of a continuous infusion of hydrocortisone 200 mg.24 h ⁻¹	

6.11 Oral contraceptives and hormone replacement therapy

These medications contain estrogen which may increase the risk of venous thromboembolism. The recommendation is to withhold combined oestrogen and progesterone oral contraceptive pills 4-6 weeks before high-risk surgery (e.g., orthopaedic) or in patients where immobility may be prolonged after surgery. Hormone replacement therapy (HRT) should be stopped 1-2 weeks before high-risk surgeries. These medications should only be restarted after at least 2 weeks of full mobilisation. Progesterone-only contraceptives need not be stopped for any surgery. Minor surgeries do not mandate stoppage of any contraceptive medications.

If discontinuation is not possible or there is not enough time before surgery, patients must be made aware of the potential risk of venous thromboembolism, and prophylaxis should be planned for the peri-operative period.

Patients should be advised to contact their general practitioner for advice regarding alternative methods of contraception during this period.

6.12 Non-prescription and herbal medication:

Non-prescription, complementary, alternative, or herbal medicines can sometimes cause problems including increased bleeding, sedation, and cardiovascular side-effects. Some examples include fish oil, garlic, ginger, ginseng, and turmeric, but this list is not exhaustive. All non-prescription medications should be stopped a week before surgery.

6.13 Medical marijuana

The preadmission Clinic Nurse will consult the anaesthetist regarding the management of patients on medical cannabis.

The anaesthetist will:

- Take a detailed history of cannabis use, including the specific product, frequency, duration, and route of administration (e.g., smoked, inhaled, or oral), and try to quantify use, such as grams of smoked cannabis or milligrams of CBD/THC from oils.
- If the patient is using a legally prescribed product, consider discussing any potential dose adjustments.
- Consider weaning if there is at least 7 days until surgery, especially if using high doses. (Avoid weaning within 24 hours: Do not attempt to wean within 24 hours of surgery, as this may precipitate withdrawal symptoms. Weaning targets: Examples of targets for weaning include <1.5g/d of smoked cannabis, <300mg/d of CBD oil, or <20mg/d of THC oil.

7. Policy context

The following documents are required to give effect to this policy:

- Legislative
 - WA Medicines and Poisons Regulations 2016
- HDWA
 - [High Risk Medications Policy](#)
 - [Consent to Treatment](#)
 - [Clinical Handover Policy](#)
 - [Medication Review Policy](#)
 - [Management of Schedule 8 and Restricted Schedule 4 Oral Liquid Medicines](#)

- [Clinical Incident Management Policy](#)
- [Code of Practice for the Handling of Schedule 8 Medicines \(drugs of addiction\) in Hospitals and Nursing Posts](#)
- East Metropolitan Health Service (EMHS)
 - [Patient Identification and Procedure Matching \(EMHS: 21\)](#)

8. Supporting information

The following documents support the implementation of this policy:

- [Medication Policy](#)
- [Medication Management Procedure](#)
- [Patient information: ANZCA Preparing for your medical procedure when taking some diabetes and/or weight loss medications](#)
- [Patient form: ANZCA Preparing for your medical procedure when taking some diabetes and/or weight loss medications](#)

9. Compliance, monitoring and evaluation

Compliance against this guideline will be evaluated and monitored by the Perioperative Management Committee. Monitoring activities will include clinical incident reported data via the Datix Clinical Incident Management System (CIMS).

10. Relevant standards

National Safety and Quality Health Service (NSQHS) Standards (second edition):

- Standard 4

11. References

1. Wood DM, Macpherson R. Australian and New Zealand guidelines for preoperative fasting. *Anesth Intensive Care* 2007; 35(4):622-623.
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8. ADS-ANZCA Perioperative Diabetes and Hyperglycaemia Guidelines, November 2022.
9. Woodcock, Tom & Barker, et al. (2020). Guidelines for the management of glucocorticoids during the peri-operative period for patients with adrenal insufficiency: Guidelines from the Association of Anaesthetists, the Royal College of Physicians and the Society for Endocrinology UK. Anaesthesia. 75.

12. Document contact

Head of Anaesthetics

Enquiries relating to this guideline may be directed to:

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13. Review

This guideline will be reviewed and evaluated as required to ensure relevance and currency. At a minimum it will be reviewed within one (1) year after first issue and at least every three (3) years thereafter.

Version	Effective from	Effective to	Amendment(s)
V4.1	07 03 2026	20 10 2028	Minor Review; change to table 7, process update for non-diabetics on SGLT2.

14. Authorisation

This guideline has been authorised and issued by:

Authorised by	Dr Tania Strickland - Director Clinical Services
Authorisation date	26 03 2026
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